

Haokai Ding

Abu Dhabi, UAE

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Summary

Robotics researcher working across underactuated manipulation, aerial robotics, and deployment-focused robot learning. Building hardware prototypes, analytical mechanisms, and reinforcement-learning systems, with a focus on understanding and closing sim-to-real gaps.

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

M.Sc. IN ROBOTICS

Abu Dhabi, UAE

Aug. 2026 – Present

Tsinghua University | Shenzhen X-Institute

TSIEN EXCELLENCE IN ENGINEERING PROGRAM (TEEP) – X SCHOLAR

Shenzhen, China

Sep. 2023 – Jul. 2026

- Joint training program in underactuated robotics and mechanism design.

Shenzhen Technology University – Future Technology School

B.ENG. IN ELECTRONIC SCIENCE AND TECHNOLOGY

Shenzhen, China

Sep. 2022 – Jul. 2026

- GPA: 85.08/100. Research focus: robotics and mechanism design. Top 10 Outstanding Graduate.

Research Experience

State Key Lab of Mechanical Systems & Vibration, Shanghai Jiao Tong University

VISITING STUDENT – UAV PERCHING AND AERIAL MANIPULATION

Shanghai, China

Feb. 2026 – Present

- Advisor: Prof. Wei Dong.
- Working on UAV perching and contact-rich aerial manipulation.

Independent Research Project (Remote GPU Workstation)

INDEPENDENT RESEARCHER – REINFORCEMENT LEARNING FOR AGILE QUADROTOR FLIGHT

Shanghai, China

Spring 2026 – Present

- Trained PPO policies with recurrent memory and staged specialization in Isaac Sim / OmniDrones for a 13-gate course containing a climbing split-S maneuver.
- Conducted a 31-factor sim-to-real audit across dynamics, sensing, and latency; identified unmodeled aerodynamic drag, observation delay, and state-estimation drift as deployment blockers.
- Built and ground-tested a deployment stack with Jetson, PX4, and Intel RealSense T265, including motion-capture-based drift characterization.

Shenzhen X-Institute, Tsinghua University

UNDERGRADUATE RESEARCHER – UNDERACTUATED ROBOTIC GRIPPERS

Shenzhen, China

2023 – 2026

- Designed and tested underactuated grippers based on semi-Peaucellier linkages.
- Developed idle-stroke and delay-triggering mechanisms; validated prototypes via simulation and physical experiments.

Publications

† Co-first author * Corresponding author

SELECTED CONFERENCE PAPERS (ROBOTICS)

[P1] A Novel Gripper with Semi-Peaucellier Linkage and Idle-Stroke Mechanism for Linear Pinching and Self-Adaptive Grasping

H. DING, W. ZHANG*

IEEE/RSJ IROS 2025

[P2] Semi-Peaucellier Linkage and Differential Mechanism for Linear Pinching and Self-Adaptive Grasping

H. DING, Z. CHEN, T. YANG*, W. ZHANG*

IEEE CASE 2025

[P3] BioCrest – A Flexible Adaptive Robotic Hand Based on Idle-Stroke Mechanism

H. DING, X. ZHONG, W. HUANG, T. YANG*, W. ZHANG*

IEEE ROBIO 2025

[P4] GLINT: An Idle-Stroke Grasp-and-Lift Hand for In-Hand Manipulation

H. DING, H. LUAN, T. YANG*, W. ZHANG*

IEEE ICCMA 2025

[P5] Peaucellier Gripper: A Novel Underactuated Gripper for Linear Pinching and Self-Adaptive Grasp

H. DING, J. FAN, Z. ZHU, Y. ZHAO, K. CHEN*, W. ZHANG*

IEEE ICARM 2025

[P6] An Underactuated Gripper with Grasshopper-Inspired Linkages and Delay Triggering for Pinching and Adaptive Grasping

H. DING[†], Y. CHEN[†], D. JIA, W. ZHANG^{*}

IEEE ICIA 2025

[P7] An Idle-Stroke Underactuated Gripper with Independent Double-Parallelogram Linkages for Pinching and Adaptive Grasping

Y. CHEN, H. LUAN, H. DING, W. ZHANG^{*}

IEEE RAAI 2025

OTHER PUBLICATIONS

[P8] Recent Progress on the Research of 3D Printing in Aqueous Zinc-Ion Batteries

Y. LIU[†], H. DING[†], H. CHEN, H. GAO, J. YU, F. MO, N. WANG^{*}

Polymers (MDPI), 2025

[P9] Rapid Analysis of Heavy-Metal Element Adsorption by SCG Based on LIBS Technology

H. XIE, L. YOU, X. FANG, L. LI, H. DING, Z. ZHOU, G. ZHANG, D. ZHANG

E3S Web of Conferences, 2025

Honors & Awards

2026	Top 10 Outstanding Graduate , Shenzhen Technology University	Shenzhen, China
2025	Undergraduate Champion , ASME Student Mechanism & Robot Design Competition	USA
2025	National Scholarship , Ministry of Education of China	China
2024	President's Award , Shenzhen Technology University	Shenzhen, China
2023	Gold Prize , 9th China "Internet+" Innovation Competition (Guangdong Division)	Guangdong, China
2023	Second Prize , 17th "Challenge Cup" Extracurricular Science Competition (Guangdong Division)	Guangdong, China

Professional Service

2025	Conference Reviewer , IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
2025	Session Chair , IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
2025	Conference Reviewer , IEEE International Conference on Automation Science and Engineering (CASE)